1. A small box with mass m is attached to a spring with spring constant k. The system is set into motion on a frictionless horizontal surface by displacing the box by a distance A from the equilibrium position and released from rest.

$V_0 = 0 \quad X_0 = A \quad \omega = \sqrt{\frac{K}{m}}$

(a.) Write down the equation of motion for the system.

~~$X(t) = A\cos(\omega t)$~~ $\quad \ddot{X} = -\omega^2 X, \quad \omega = \sqrt{\frac{K}{m}}$

$X = X_0\cos(\omega t) = A\cos(\omega t)$

(b.) Find the initial total energy of the system.

$E_i = T_i + U_i = \frac{1}{2}KX^2 = \frac{1}{2}KX(t=0)^2 = \frac{1}{2}KA^2$

(c.) Find an expression for the potential energy as a function of position x.

$U = \frac{1}{2}KX^2 = \frac{1}{2}KA^2\cos^2(\omega t)$

(d.) Using the initial total energy and your expression for potential energy, find an expression for the kinetic energy as a function of position x only.

$\frac{1}{2}KA^2 = T + \frac{1}{2}KX^2, \quad T = \frac{1}{2}K(A^2 - X^2)$

(e.) At what displacement are the potential energy and kinetic energy equal?

$\frac{1}{2}K(A^2 - X^2) = \frac{1}{2}KX^2$

$2X^2 = A^2, \quad X = \frac{A}{\sqrt{2}}$

2. Utilizing the same spring and mass as the previous problem, the mass now undergoes a damping force proportional to its velocity, $F_d = -bv$.

(a.) Write down the modified *differential* equation of motion for the system, in terms of x (along with its derivatives), β, and ω_0.

$$m\ddot{x} + b\dot{x} + Kx = 0 \quad , \quad \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$

$$2\beta = \frac{b}{m}, \quad \omega_0 = \sqrt{\frac{K}{m}}$$

(b.) Write down the general solution for the case that $\beta < \omega_0$.

$$x(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

(c.) What is the expression for the damped angular frequency in this case?

$$\omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

(d.) If $m = 1kg$, $k = 6N/m$, and $b = 2\sqrt{2}kg/s$, what is the value of the damped angular frequency?

$$\omega_1 = \sqrt{\frac{K}{m} - \left(\frac{b}{2m}\right)^2}$$

$$= \sqrt{\left(\frac{6}{1}\right) - \left(\frac{2\sqrt{2}}{2(1)}\right)^2} = \sqrt{4} = 2\ Hz$$